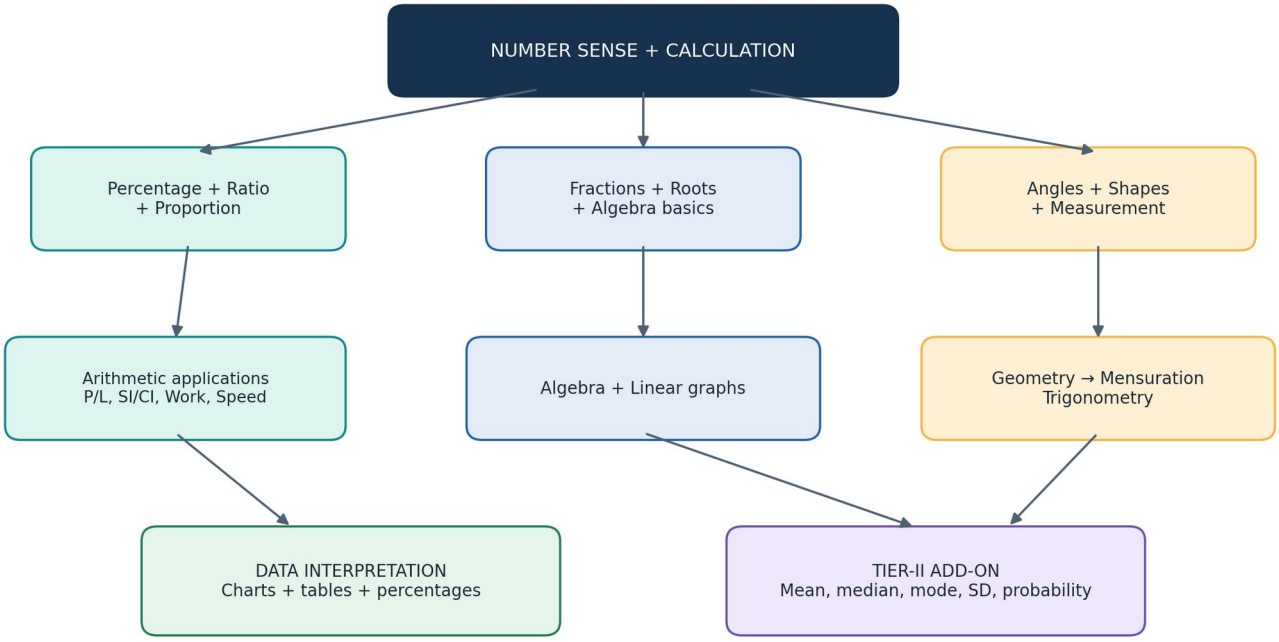
SSC CGL

# MATHEMATICS MASTER SYLLABUS

## Tier-I Quantitative Aptitude + Tier-II Mathematical Abilities

Official syllabus mapping • Historical topic priorities • Prerequisite learning map • 24-week plan • Reusable checklists




**THE DESIGN PRINCIPLE**

Study in dependency order, allocate time using historical evidence, and measure mastery using fresh questions. Do not confuse third-party frequency analysis with official SSC weightage.

| TRACK | REFERENCE |
|---|---|
| Tier-I | 25 Q \| 50 marks \| 15 min |
| Tier-II Paper-I Maths | 30 Q \| 90 marks \| 30 min |
| Core syllabus | Shared foundation, Tier-II add-ons |
| Study horizon | 24-week reusable plan |

*Use this guide as a syllabus checklist and revision dashboard. It is not an official SSC publication. The latest notification takes precedence if rules change.*

# 1. Exam blueprint: what the two tiers require

The topic universe overlaps heavily, but the exam setting differs. Tier-I rewards quick, dependable decisions; Tier-II uses a longer mathematics sectional timer but applies a larger penalty for incorrect answers.

| Item | Tier-I: Quantitative Aptitude | Tier-II: Mathematical Abilities (Paper-I) |
|---|---|---|
| Questions | 25 | 30 |
| Maximum marks | 50 | 90 |
| Subject timer | 15 minutes | 30 minutes |
| Marks per correct answer | +2 | +3 |
| Penalty per wrong answer | −0.50 | −1 |
| Official level note | Quantitative Aptitude is the Tier-I section. | Mathematical Abilities specified at Matriculation level in the 2026 notice. |

## How the syllabus is divided

| Shared core — study for both tiers | Explicit Tier-II emphasis | Separate post-specific papers |
|---|---|---|
| Number systems; arithmetic; algebra; linear-equation graphs; geometry; mensuration; trigonometry; tables and charts. | Simple + compound interest; mean, median, mode, standard deviation; simple probability; stronger mixed application. | Paper-II Statistics is for eligible/shortlisted JSO and Statistical Investigator candidates. Paper-III Finance & Economics is for eligible/shortlisted AAO/AAO candidates. |

**Important scope distinction**

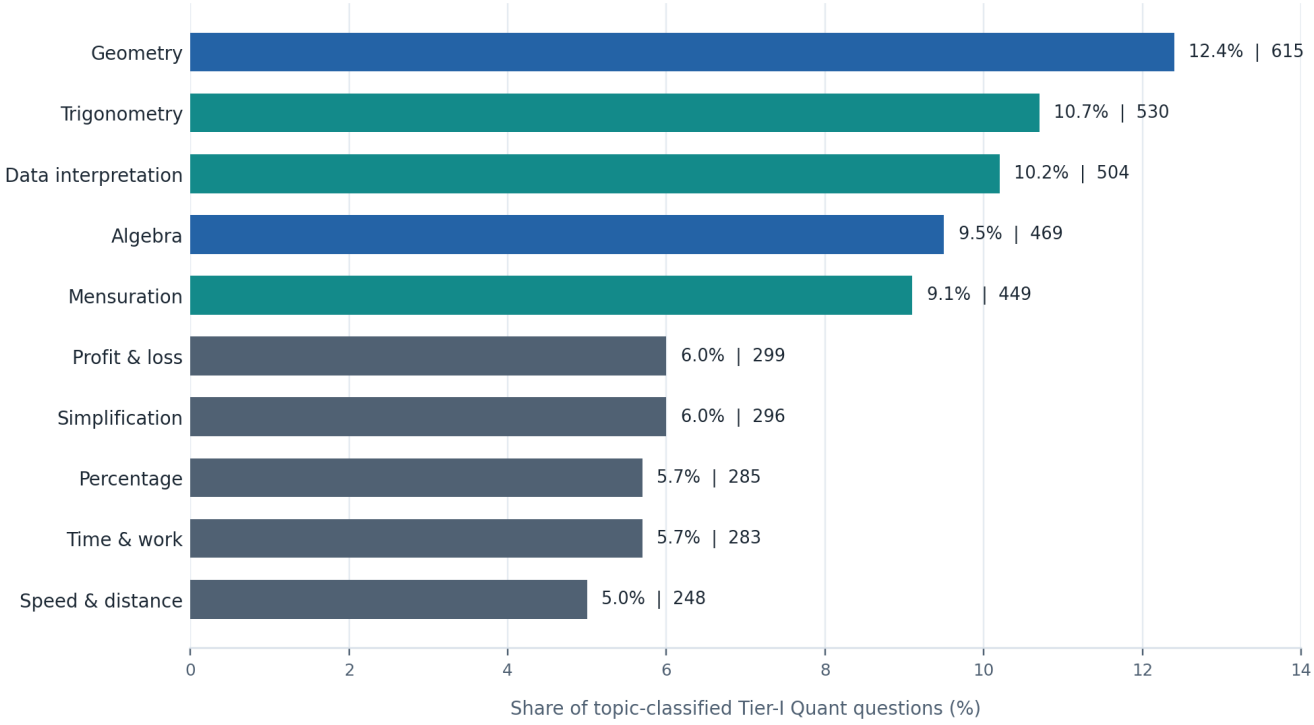
For most candidates, “Tier-II maths” means Mathematical Abilities in compulsory Paper-I. The separate Paper-II Statistics is not an extra section all candidates take. Check the post and eligibility conditions in the notification.

## Official rule references

- SSC CGL 2026 notice, Section 13.8: Tier-I scheme and negative marking.
- SSC CGL 2026 notice, Section 13.9: Tier-II Paper-I structure and penalty.
- SSC CGL 2026 notice, Sections 13.10.3 and 13.11.1: Tier-I Quant and Tier-II Mathematical Abilities scope.

# 2. Historical paper evidence: where to allocate attention

A third-party analysis reports 4,958 topic-classified Quant questions from 204 SSC CGL Tier-I papers from 2019–2025. Its top five topics account for approximately 51.8% of that classified set. The sample is useful for prioritisation, not for predicting an exact future paper.




| Topic | Count (2019–25) | Share |
|---|---|---|
| Geometry | 615 | 12.4% |
| Trigonometry | 530 | 10.7% |
| Data Interpretation | 504 | 10.2% |
| Algebra | 469 | 9.5% |
| Mensuration | 449 | 9.1% |
| Profit & Loss | 299 | 6.0% |
| Simplification | 296 | 6.0% |
| Percentage | 285 | 5.7% |
| Time & Work | 283 | 5.7% |
| Speed, Time & Distance | 248 | 5.0% |

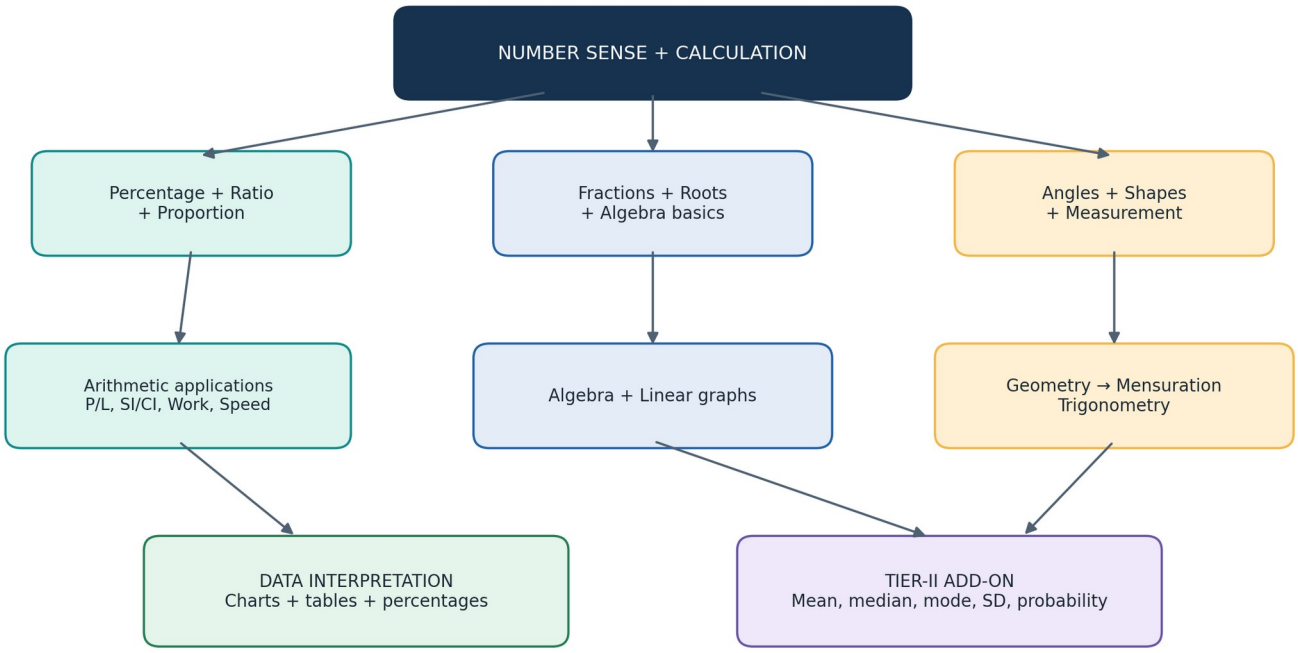
**How to read this chart**

Counts depend on which shifts were captured and how each question was tagged. SSC does not publish a fixed topic-wise weightage. Do not infer that a given topic is guaranteed to appear.

*Source: Comperly Acad, "SSC CGL Topic-Wise Weightage: What 204 Real Papers From 2019–2025 Show," updated 6 September 2026. Link in the Sources section.*

# 3. The dependency map: learn concepts in a smart order

A prerequisite map prevents wasted effort. Percentages and ratios feed many arithmetic chapters; geometry is the base for mensuration; computation powers Data Interpretation (DI).




## Recommended learning order

| Stage | Learn | Why it comes here |
|---|---|---|
| 1. Fluency | Whole numbers, fractions, decimals, ratios, roots, calculation habits | Removes calculation bottlenecks across the syllabus. |
| 2. Arithmetic engine | Percentage → ratio → average → profit/loss → interest → mixture/partnership → work → speed | Builds on transferable relationships rather than isolated tricks. |
| 3. Algebra + graphs | Identities, factorisation, surds, linear equations and graphs | Develops symbolic fluency and equation interpretation. |
| 4. Geometry → mensuration | Lines/angles → triangles → circles → quadrilaterals/polygons → areas/solids | Theorems and diagrams support measurement questions. |
| 5. Trigonometry | Ratios → standard values → identities → heights/distances | Formula recall is easier after geometry and triangle basics. |
| 6. DI + Tier-II statistics | Tables/graphs → average/percentage calculations → mean/median/mode/SD/simple probability | Applies arithmetic and adds explicit Tier-II coverage. |
| 7. Mixed execution | PYQs, sectional tests, error review, full simulations | Checks transfer: solving correctly when the topic is not named. |

**Study rule**

After the first pass, stop practising only chapter-labelled sets. Use mixed sets so you must identify the method yourself, as in the real exam.

# 4. Master checklist A: number sense + arithmetic foundations

Tick a box only after you can explain the method, solve fresh questions without notes, and correct your own mistakes.

## A. Number systems and calculation — P0 foundation

[ ] Whole numbers, integers, number relationships and ordering.

[ ] Fractions and decimals; conversion between fraction, decimal and percentage forms.

[ ] Operations on fractions/decimals; order of operations and simplification.

[ ] Square roots, squares/cubes needed for exam calculation; elementary surds as specified in algebra.

[ ] Divisibility, prime factors, HCF and LCM as foundational number-sense practice.

[ ] Calculation drills: multiplication tables, fraction comparisons, squares and common percentage equivalents.

## B. Percentage, ratio, proportion and average — P0/P2

| Topic | Concept checklist | Common use |
|---|---|---|
| Percentage | Find x% of a value; reverse percentage; increase/decrease; successive change; percentage comparison. | Profit/loss, interest, DI, comparison questions. |
| Ratio & proportion | Equivalent ratios; divide a total in a ratio; direct/inverse proportion; compare ratios. | Mixture, partnership, work rates and chart questions. |
| Average | Arithmetic mean; missing value; replacement; combined/weighted average. | Data interpretation, ages and group-based arithmetic. |

## C. Calculation mastery gate

| Check | Ready standard | My status |
|---|---|---|
| Fresh basic set | At least 18/20 correct, with errors understood. | [ ] Not yet [ ] Done |
| No-calculator calculation | Can compare fractions, percentages and ratios without lengthy working. | [ ] Not yet [ ] Done |
| Mixed recall | Can choose between percentage, ratio and average without a chapter label. | [ ] Not yet [ ] Done |

# 5. Master checklist B: arithmetic applications + algebra

## D. Arithmetic applications — P2, built on foundation

| Chapter | Concept checklist |
|---|---|
| Interest | Simple interest; compound interest; annual compounding; amount; principal/rate/time; compare SI and CI. |
| Profit, loss and discount | Cost price, selling price, marked price; profit/loss percent; discount; successive discounts; reverse-price problems. |
| Mixture and alligation | Mixing ratios; concentration; replacing/removing a quantity; finding missing amounts. |
| Partnership | Capital × time; ratio of investment; changing investments/duration; dividing profits. |
| Time and work | Work rate; combined work; efficiency; worker-day relationships; wage sharing; pipes/cisterns grouped as rate problems. |
| Time, speed and distance | Unit conversions; speed/time/distance; relative speed; trains; boats and streams; average speed. |

## E. Algebra and linear-equation graphs — P1

[ ] Basic algebraic identities from school algebra; expansion and factorisation.

[ ] Simplifying algebraic expressions; substitution and elementary surds.

[ ] Linear equations and word problems that convert into equations.

[ ] Graphs of linear equations: axes, coordinates, ordered pairs, plotting points and reading a straight-line graph.

[ ] Use algebra to shorten arithmetic work rather than memorising an unrelated shortcut for each question.

**Efficient checkpoint**

Before moving on, solve a mixed set covering profit/loss, interest, work and speed. If you only solve them correctly when the chapter title is visible, the concept is not yet exam-ready.

# 6. Master checklist C: geometry — a high-priority scoring cluster

The official syllabus explicitly names triangle centres, congruence/similarity, circle chords/tangents/angles and common tangents. Treat this as a theorem-and-diagram subject, not a formula memorisation subject.

## F. Geometry

| Area | Checklist of concepts |
|---|---|
| Lines and angles | Angle pairs; parallel lines and transversal relationships; complementary/supplementary angles; angle chasing. |
| Triangles | Types and angle sums; exterior angle; triangle inequality basics; congruence criteria; similarity; proportional sides; area ratios. |
| Triangle centres | Centroid, incentre, circumcentre, orthocentre; where each is located in common triangle types; their defining properties. |
| Quadrilaterals | Parallelogram, rectangle, square, rhombus, trapezium; side, angle, diagonal and area properties. |
| Regular polygons | Interior/exterior angles; angle sums; number of sides from angle information. |
| Circles | Radius/diameter/chord; perpendicular from centre to chord; angles subtended by chords; cyclic quadrilaterals where relevant. |
| Tangents | Tangent-radius perpendicularity; equal tangents from an external point; common tangents to two or more circles. |

## Geometry solving protocol

| Step | Action |
|---|---|
| 1 | Draw or clean up the figure; mark all known lengths and angles. |
| 2 | Name the theorem or relationship before inserting numbers. |
| 3 | Check whether the problem needs a ratio, angle sum, congruence or similarity. |
| 4 | Re-check units and whether the final value actually answers the question. |

**Geometry mastery gate**

Maintain a one-page theorem sheet, but practise applying each theorem to new diagrams. A remembered formula without diagram recognition is not mastery.

# 7. Master checklist D: mensuration + trigonometry

## G. Mensuration — P1

| Shape group | Checklist |
|---|---|
| 2D: triangle, quadrilaterals, regular polygons, circle | Perimeter/circumference; area; missing dimensions; composite figures; unit conversion. |
| Prism and cuboid / rectangular parallelepiped | Cross-sectional idea; volume; lateral/total surface area; face, edge and dimension reading. |
| Right circular cylinder | Radius/diameter; volume; curved surface area; total surface area. |
| Right circular cone | Radius, height, slant height; volume; curved and total surface area. |
| Sphere and hemisphere | Surface area and volume; curved versus total area for a hemisphere. |
| Regular right pyramid (triangular/square base) | Base area; vertical height vs slant height where applicable; volume/surface-area interpretation. |

## H. Trigonometry and heights/distances — P1

[ ] Sine, cosine, tangent and reciprocal ratios; connect ratio definitions to a right triangle.

[ ] Standard values for common angles; degree and radian measures.

[ ] Complementary-angle relationships and standard identities (including $\sin^2\theta + \cos^2\theta = 1$).

[ ] Height and distance: angles of elevation/depression, vertical objects, horizontal distance, drawing a right-triangle model.

[ ] Choose the appropriate ratio from the sides provided; do not memorise a shortcut without its conditions.

**Frequent error to prevent**

In mensuration, distinguish radius from diameter and curved surface area from total surface area. In height-and-distance problems, sketch the triangle before choosing the ratio.

# 8. Master checklist E: data interpretation + Tier-II statistics

## I. Tables, charts and graphs — shared visual-data skill

| Display type | What to practise |
|---|---|
| Table | Totals, ratios, averages, rank/order, percentage share and changes between rows/columns. |
| Bar diagram | Read axes and scale; compare categories; calculate differences and percentage changes. |
| Pie chart | Translate angle/share into a quantity; compare sectors; use totals and percentages accurately. |
| Histogram | Interpret grouped intervals and frequencies; read the bars correctly. |
| Frequency polygon | Read class intervals, frequencies and trend patterns. |

## J. Explicit Tier-II Mathematical Abilities add-on

[ ] Mean (arithmetic average), median (middle value after sorting) and mode (most frequent value).

[ ] Standard deviation: understand spread around the mean and solve the level of calculation appropriate to the exam.

[ ] Simple probability: outcomes, favourable outcomes, total equally likely outcomes and straightforward event calculations.

[ ] Interpret statistics displayed in tables, histograms, frequency polygons, bar diagrams and pie charts.

**Tier boundary**

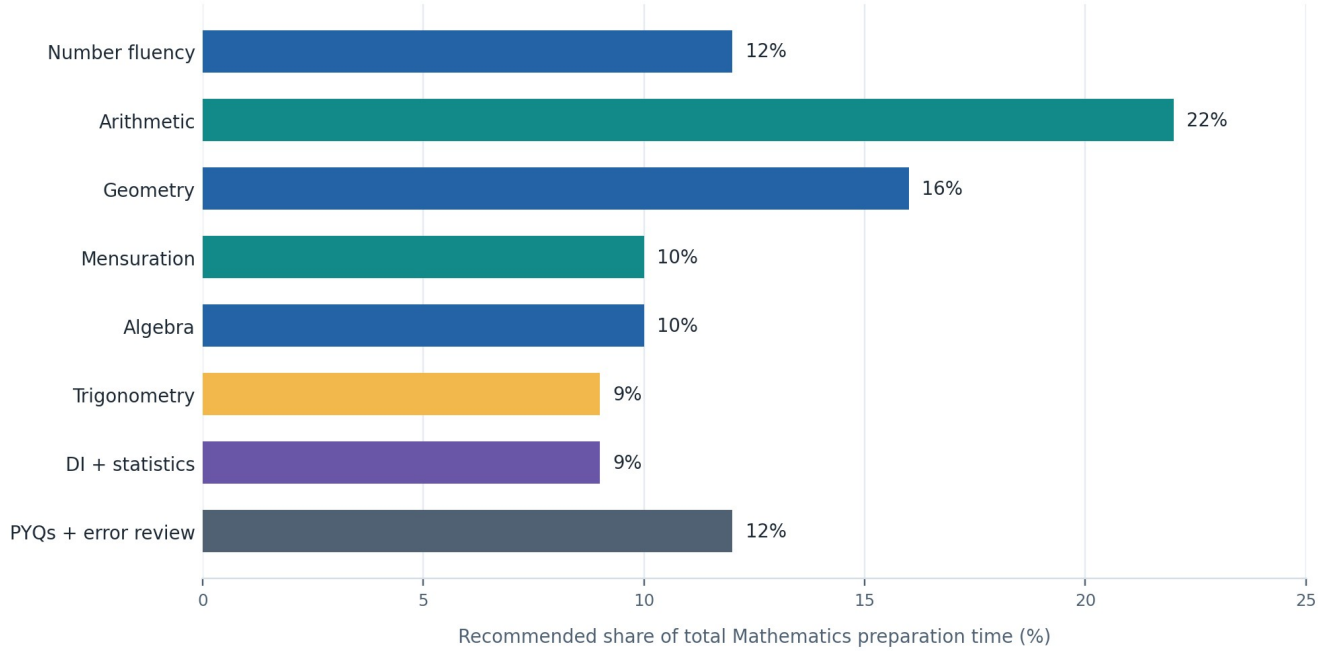
The 2026 notice explicitly includes mean, median, mode, standard deviation and simple probability under Tier-II Mathematical Abilities. It is safest to learn these for Tier-II even if your Tier-I practice focuses more heavily on reading charts.

## Separate route: Paper-II Statistics

Only for candidates applying to and shortlisted for JSO / Statistical Investigator posts: Paper-II Statistics is a separate 100-question paper (200 marks, 2 hours in the 2026 notice). Its syllabus is substantially broader than Mathematical Abilities and includes data collection/presentation, measures of central tendency and dispersion, moments/skewness/kurtosis, correlation/regression, probability theory and random variables/distributions. Consult the separate Paper-II syllabus in the official notice if this route applies to you.

# 9. Allocate study time by evidence and learning cost

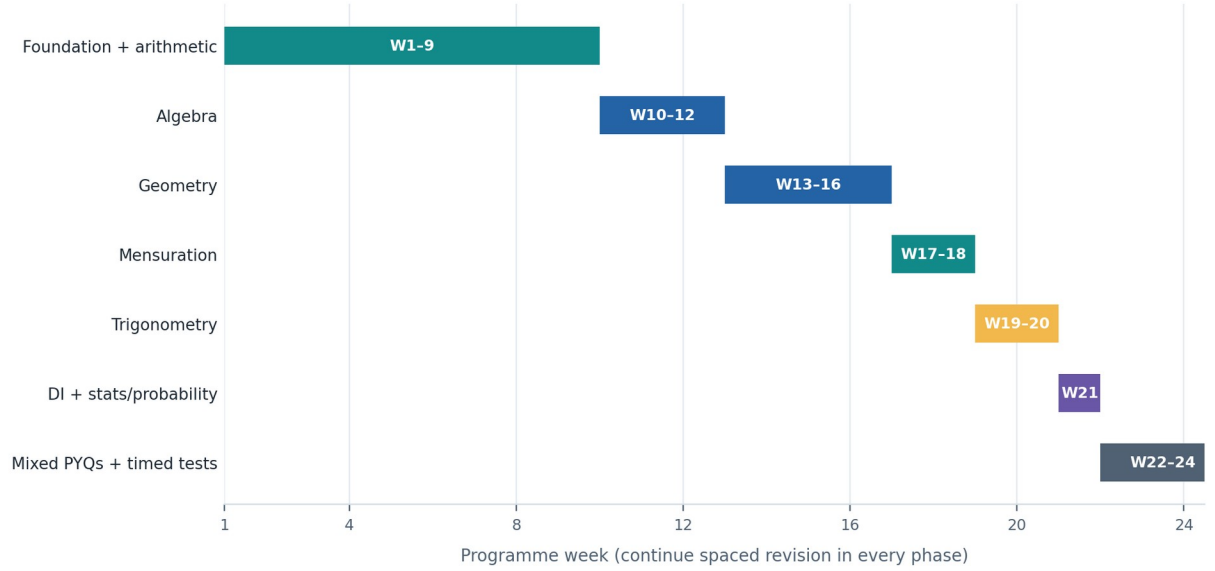
The percentages below are recommended shares of your total Mathematics preparation time, not official marks weightage and not a prediction of the paper.




| Area | Study-time share | Reason |
|---|---|---|
| Arithmetic | 22% | Large transfer value across independent chapters and DI. |
| Geometry | 16% | Largest topic in the cited historical dataset. |
| Number fluency | 12% | Base skill that improves speed and lowers errors. |
| PYQs + error review | 12% | Converts knowledge into exam execution. |
| Mensuration | 10% | High-frequency topic; depends on geometry and formula choice. |
| Algebra | 10% | Strong historical representation; supports symbolic reasoning. |
| Trigonometry | 9% | High-frequency and formula/relationship heavy. |
| DI + statistics | 9% | Important historical DI count and explicit Tier-II add-ons. |

# 10. Detailed 24-week implementation plan

*The timeline shows the primary emphasis; brief recall, error review and mixed practice continue in every phase.*




| Weeks | Focus | Minimum weekly output |
|---|---|---|
| 1–2 | Number sense, fraction/decimal fluency, roots, divisibility, HCF/LCM; baseline diagnostic. | 2 short concept tests + error log started. |
| 3–4 | Percentage, ratio/proportion, average. | Chapter practice + first topic-wise PYQs. |
| 5–6 | Profit/loss/discount; simple/compound interest. | Mixed arithmetic set with reverse/successive cases. |
| 7–9 | Mixture/alligation, partnership, time/work, speed/distance. | One mixed arithmetic test each week. |
| 10–12 | Algebra identities, factorisation, surds, linear equations and graphs. | Timed algebra set; correct every error. |
| 13–16 | Geometry: lines, triangles/centres, congruence/similarity, circles/tangents, quadrilaterals/polygons. | Theorem sheet + diagram-based PYQs. |
| 17–18 | 2D + 3D mensuration: areas, perimeter, surface area, volume. | Formula map + mixed solids test. |
| 19–20 | Trigonometric ratios, standard values, identities, complementary angles, heights/distances. | Formula recall + applied questions. |
| 21 | Tables/graphs, DI, mean/median/mode, standard deviation, simple probability. | Tier-II add-on checklist complete. |
| 22 | Mixed PYQs across every chapter. | Tag mistakes C/M/K/T and choose top 3 weak areas. |
| 23 | Tier-I 15-minute and Tier-II 30-minute mathematics simulations. | At least 2 timed sets with post-test analysis. |
| 24 | Repair weak concepts, revise formulas/theorems, full sectional simulations. | Final mastery audit and next-cycle plan. |

### Time-fit adjustment

If a chapter fails its mastery gate, extend it and compress low-value rewatching—not PYQ analysis or review. Progress should be based on your score and error pattern, not calendar pressure.

# 11. Daily operating system + mastery gates

## A. 100-minute daily Mathematics block

| Minutes | Task | What counts as done |
|---|---|---|
| 10 | Spaced recall | Recall formulas/theorems from memory; revisit yesterday's error. |
| 25 | Learn one concept | One focused lesson plus short notes, not a long copying session. |
| 30 | Accuracy practice | Fresh questions; solve before reading solutions. |
| 25 | PYQs / mixed practice | Use authentic previous-year questions where possible; record year/shift/source. |
| 10 | Error analysis | Classify the error and re-solve without looking. |

## B. Four error categories

| Code | Diagnosis | Repair action |
|---|---|---|
| C — Concept | Did not know which rule/theorem applied. | Relearn the concept, then solve 5–10 new examples. |
| M — Method | Knew the idea but chose a slow/unsuitable approach. | Compare methods and practise recognising the shortest valid one. |
| K — Calculation | Sign, fraction, arithmetic or unit error. | Run targeted computation drills and write the exact slip. |
| T — Time | Correct solution, but too slow for the test. | Practise similar questions in short timed sets; build skip/return discipline. |

## C. Mastery gates

| Gate | Suggested criterion | Status |
|---|---|---|
| Concept clarity | At least 18/20 fresh basic questions correct. | [ ] |
| Chapter mastery | 85–90% correct in a fresh mixed chapter set; understand every miss. | [ ] |
| Tier-I execution | Build towards 23–25 correct out of 25 in 15 minutes. Stretch target, not a cutoff. | [ ] |
| Tier-II execution | Build towards 27+ correct out of 30 in 30 minutes, with disciplined selection. Stretch target. | [ ] |
| Spaced revision | Re-solve key errors after about 1 day, 1 week and 3 weeks. | [ ] |

# 12. Reusable chapter progress tracker

Print this page or duplicate the checklist in your own notes. A chapter is “done” only after the concept, questions and error review are all completed.

| Topic | Priority | Concept | Questions | PYQ | Review |
|---|---|---|---|---|---|
| Number sense / computation | P0 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Percentage | P0 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Ratio and proportion | P0 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Averages | P0 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Profit, loss and discount | P2 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Simple and compound interest | P2 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Mixture and alligation | P2 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Partnership | P2 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Time and work | P2 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Time, speed and distance | P2 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Algebra identities / factorisation / surds | P1 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Linear equations and graphs | P1 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Lines and angles / triangles / centres | P1 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Congruence and similarity | P1 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Circles / chords / tangents | P1 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Quadrilaterals and regular polygons | P1 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| 2D mensuration | P1 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| 3D mensuration | P1 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Trigonometric ratios / standard values | P1 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Identities / complementary angles / heights | P1 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Tables / bar / pie / histogram / frequency polygon | P1 | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Mean / median / mode | Tier-II add-on | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Standard deviation | Tier-II add-on | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Simple probability | Tier-II add-on | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |
| Mixed PYQ sets + full sectional timing | Ongoing | [ ] Learn | [ ] Practise | [ ] PYQ | [ ] Review |

## Personal benchmark log

| Date | Set / topic | Correct / attempted | Time taken | Main error type | Next review |
|---|---|---|---|---|---|
| | | | | | |
| | | | | | |
| | | | | | |
| | | | | | |

# 13. Reliable resources and references

## Official source first

1. SSC CGL 2026 official notification — syllabus and exam scheme. Use Sections 13.8–13.11 for Tier-I, Tier-II Paper-I Mathematics Abilities and post-specific papers.

https://ssc.gov.in/api/attachment/uploads/masterData/NoticeBoards/Notice_of_adv_cgl_2026.pdf

## Historical question pattern

2. Comperly Acad, SSC CGL syllabus and quantified topic analysis (204 papers, 2019–2025; 4,958 tagged Quant questions). Third-party analysis; methodology and topic tags are theirs.

https://acad.comperly.com/ssc-cgl/syllabus

3. SSC CGL topic-wise weightage analysis across 204 papers:

https://acad.comperly.com/insights/ssc-cgl-topic-wise-weightage

## Practice resources

**4. Official SSC portal / announcements / answer keys**

https://ssc.gov.in/

**5. Topic-wise Quantitative Aptitude PYQs (third-party bank)**

https://www.unlockias.in/ssc-cgl-previous-year-questions/quantitative-aptitude

**6. Tier-I previous paper archive**

https://testbook.com/ssc-cgl-exam/previous-year-papers?stage=Tier+I&year=2025

**7. NCERT ePathshala for foundational concept repair**

https://epathshala.nic.in/

**8. Maths Foundation Batch for All Exams — Aditya Sir playlist**

https://www.youtube.com/playlist?list=PLXTyt_wUBqQ5Axt76rasrdgZ4SYE27SCe

## Source limitations and use

- Official SSC notice defines the syllabus and marking scheme; it does not publish fixed topic-wise question weightage.
- The historical chart is a third-party classification of 2019–2025 papers. Treat it as a planning signal, not a guarantee.
- PYQ websites can differ in transcription and tagging. Prefer original SSC response sheets and official answer keys when available.
- Re-check the notification for the actual recruitment cycle before the exam; dates and rules may change.

> **Final principle**
>
> Finish the syllabus once, then keep improving the quality of your attempts: recognition speed, calculation accuracy, choice of method, and error reduction.